AURA Twin

History + wearable state.
Fresh-patient evidence.
A challengeable prediction.

AURA-SYNTHETIC-3.0.0

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Generated data only · No clinical validation or treatment recommendation

Freeze the prediction. Then reveal the outcome.

Current model aura-synthetic-3.0.0 · synthetic benchmark evidence

01

Choose

Patient, cutoff, sensor case.
Copy and truncate history.

02

Freeze

Predict before seeing outcomes.
Preserve output + hashes.

03

Reveal

Original recorded future.
No recomputation.

04

Compare

Model vs persistence.
Interval hits + receipt.

Public data: this enforces a computation boundary, not secure study blinding.

Four sensor cases expose different failure modes.

Current model aura-synthetic-3.0.0 · synthetic benchmark evidence

Clean replay

An honest baseline; a local window may lose to persistence.

10-minute CGM gap

Forward fill permitted; show degraded quality.

30-minute dropout

Stale signal; withhold forecast and probability.

Undetected sensor bias

Plausible +30 mg/dL input shift may pass freshness checks.

The reference future stays unchanged. Persistence sees the same perturbed current value.

Compare fixed policies on identical new windows.

Current model aura-synthetic-3.0.0 · synthetic benchmark evidence

Routine	FN: v2 → v3	FP: v2 → v3	V3 recall	V3 precision
Regular	211 → 212	1,207 → 1,134	82.57%	46.96%
Meals +2h	164 → 159	1,366 → 1,370	85.64%	40.90%
Meals +4h	132 → 142	1,332 → 1,195	84.58%	39.46%
Jittered meals	491 → 452	1,541 → 1,384	65.26%	38.02%
Daily changes + skips	324 → 313	1,627 → 1,490	69.13%	31.99%
Variable sleep/activity	420 → 391	1,657 → 1,597	60.06%	26.91%

Withheld combination: missed positives 324 → 330 · false positives 1,657 → 1,624

Withheld combination is separate from this six-family primary table.

Make the experiment reproducible offline.

Current model aura-synthetic-3.0.0 · synthetic benchmark evidence

01

What can be checked

Raw model bytes → SHA-256
Canonical input prefix → SHA-256
Recomputed forecast and revealed score
Mismatch detection for altered artifacts

02

What is not established

No digital signature
No independent timestamp
No trusted authorship guarantee
No clinical audit certification

```
node scripts/verify_receipt.mjs path/to/receipt.json
```

A forecast you can inspect.

A working fusion pipeline.
A measured synthetic comparison.
A reproducible challenge.

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